ADITHYA BHAT

Visa Research, Visa Inc., CA

◇ Email - dth.bht@gmail.com ◇ GitHub - https://github.com/libdist-rs

◇ Website - https://adithyabhatkajake.github.io

SUMMARY

Security-focused systems engineer and researcher with 8+ years of experience building secure, fault-tolerant distributed systems and cryptographic protocols. Published 13 peer-reviewed papers at top security venues (IEEE S&P, CCS, NDSS) and hold 3 US patents. Deep expertise in applied cryptography, secure protocol design, threat modeling, and privacy-enhancing technologies. Experienced in Go, Rust, C/C++, and Python across production systems and research prototypes.

TECHNICAL SKILLS

Languages	Python, Go, Rust, C/C++, TypeScript
Security & Crypto	Cryptographic protocol design, Key lifecycle management, Threat modeling, Secure MPC, PVSS, Threshold signatures, PKI
Systems & Infra	Fault-tolerant distributed systems, Supply chain security, Secure infrastructure design, Authentication/Authorization
Tools & Platforms	Docker, Git, Linux, gRPC, Protobuf, Ethereum, libp2p

WORK EXPERIENCE

Visa Research, Foster City (Aug 2023 - Present)

Staff Research Scientist

- Designed and built cryptographic protocols for Visa's payments infrastructure, including an accountable decentralized anonymous payment system (US patent filed) ensuring transaction privacy while maintaining auditability.
- Led research on fully homomorphic encryption applied to smart contracts (SoK accepted at DeFi 2026), evaluating the security and feasibility of privacy-preserving computation on untrusted platforms.
- Developed FastSync, a secure blockchain synchronization protocol using compact cryptographic proofs to verify state integrity without replaying full transaction history. Filed US patent.
- Architected a sharding protocol with secure cross-shard transaction processing, addressing consistency and integrity guarantees in partitioned distributed systems.
- Co-authored research on private set intersection protocols, enabling privacy-preserving data matching across organizations without revealing underlying datasets.

VMware Research, Remote (May 2021 - Aug 2021)

Research Intern

- Built a privacy-preserving anonymous token system (UTT) on VMware's Concord-BFT platform, enabling decentralized ecash with accountable privacy—users transact anonymously while authorities can audit when needed.
- Designed Quick-Pay, a low-latency payment protocol with a lock-free sharding extension, eliminating coordination bottlenecks while maintaining security invariants in distributed transaction processing.

Purdue University, West Lafayette (Aug 2018 - Aug 2023)

Graduate Research Assistant – Department of Computer Science

- Designed and implemented secure distributed protocols (OptRand, RandPiper) for cryptographic randomness generation, providing verifiable, unpredictable, and bias-resistant random beacons for security-critical applications.
- Co-authored attack analysis and security improvements for the Tor directory protocol (IEEE S\P 2024), identifying vulnerabilities in a real-world anonymity system and proposing mitigations.
- Developed distributed key generation protocols without broadcast (Journal of Cryptology 2024), enabling secure key lifecycle management in settings where broadcast channels are untrusted or unavailable.
- Built libcrypto-rs and libnet-rs: open-source Rust libraries for cryptographic primitives and secure networking in distributed systems.
- Published 10 papers at top security venues including CCS (x4), IEEE S\P, NDSS, and Financial Cryptography. Served as PC member for CCS, USENIX Security, and FC.

EDUCATION

Ph.D., Computer Science – Purdue University, West Lafayette *(2018-2023)*
Advisor: Aniket Kate. Focus: Byzantine fault-tolerant distributed systems and applied cryptography.

B.Tech., Information Technology – NIT Karnataka, Surathkal *(2014-2018)*

SELECTED PUBLICATIONS & PATENTS

13 peer-reviewed publications at top security venues. 3 US patents. Selected highlights:

- **Attacking and Improving the Tor Directory Protocol** (IEEE S\P 2024). Identified vulnerabilities in Tor's directory system and proposed security improvements.
- **Synchronous DKG without Broadcasts** (Journal of Cryptology 2024). Secure distributed key generation without trusting broadcast channels.
- **HashRand – Efficient Asynchronous Random Beacon without Threshold Cryptographic Setup** (CCS 2024). Secure randomness generation without trusted setup ceremonies.
- **Lite-PoT – Practical Powers-of-Tau Setup Ceremony** (CCS 2025). Practical secure setup for cryptographic systems used in production.
- **Accountable Decentralized Anonymous Payments** – US Patent 20240265373. Privacy-preserving payment system with auditability.

OPEN SOURCE & SOFTWARE ARTIFACTS

- **libcrypto-rs** (Rust) – Cryptographic primitives library with traits and wrappers for threshold signatures, key management, and secure protocol building blocks.
- **libnet-rs** (Rust) – Secure networking library for distributed systems. Powers multiple BFT protocol implementations.
- **libmempool-rs** (Rust) – Transaction mempool with ordering and integrity guarantees for distributed systems.